this is my unit, kan u explain what i should multiply it with? $cm^3 \cdot mm \cdot m^{-2} \cdot d^{-1} \cdot kPa^{-1}$, 1. cm3 à cm3 (·1) 2. mm à µm (·1) 3. m-2 à m-2 (·10-4) 4. d-1 à d-1 (·1/86400) 5. kPa-1àbar-1

26 Feb

There are a few errors in your steps. Let me go through each one carefully.

Your target Barrer unit is:

10^{-10} cm^3(STP) · cm · cm^{-2} · s^{-1} · $cmHg^{-1}$

So you need to convert every dimension to match that. Let's check yours one by one:

Step 1: cm^3 → cm^3 (×1) ✅ Correct, volume is already in cm^3.

Step 2: mm → µm (×1) ❌ This is wrong in two ways. First, the **target is cm**, not µm — you need to convert mm to **cm**, not µm. Second, 1 mm ≠ 1 µm — they are completely different units (1 mm = 1000 µm).

Correct: mm → cm, × 0.1 (because 1 mm = 0.1 cm)

Step 3: m^{-2} → m^{-2} ($\times 10^{-4}$) ❌ You wrote $m^{-2} \rightarrow m^{-2}$, meaning no conversion at all — but then applied $\times 10^{-4}$, which contradicts itself. The actual conversion is **m^{-2} → cm^{-2}**, because the Barrer uses cm^{-2} for area.

Correct: $m^{-2} \rightarrow cm^{-2}$, $\times 10^{-4}$ (because 1 m^2 = 10^4 cm^2, so 1 m^{-2} = 10^{-4} cm^{-2})

Step 4: $d^{-1} \rightarrow d^{-1}$ (×1/86400) ❌ Same issue — you wrote $d^{-1} \rightarrow d^{-1}$ but the Barrer uses **s^{-1}**, not d^{-1}. The conversion is days to seconds.

Correct: $d^{-1} \rightarrow s^{-1}$, × 1/86400 (because 1 day = 86400 s)

Step 5: $kPa^{-1} \rightarrow bar^{-1}$ ❌ Wrong target unit. The Barrer uses **$cmHg^{-1}$**, not bar^{-1}. You need to go all the way to cmHg.

Correct: $kPa^{-1} \rightarrow cmHg^{-1}$, × 1/7.5006 (because 1 kPa = 7.5006 cmHg)

Then divide by 10^{-10} (the Barrer definition).

Corrected full calculation:

Step	Conversion	Factor
1	$cm^3 \rightarrow cm^3$	×1
2	mm → **cm**	×0.1
3	$m^{-2} \rightarrow$ **cm^{-2}**	$\times 10^{-4}$
4	$d^{-1} \rightarrow$ **s^{-1}**	×1/86400
5	$kPa^{-1} \rightarrow$ **$cmHg^{-1}$**	×1/7.5006

Step	Conversion	Factor
Combined	$1 \times 0.1 \times 10^{-4} \times (1/86400) \times (1/7.5006)$ =	**1.543×10^{-11}**
$\div 10^{-10}$	Barrer definition	**= 0.1543 Barrer**

So **1 $cm^3 \cdot mm \cdot m^{-2} \cdot d^{-1} \cdot kPa^{-1}$ = 0.1543 Barrer**, and to convert your permeability values you multiply by **0.1543**.

no i dont want to go to barrer anymore

26 Feb

Got it! So what do you want to convert to? And is the unit $cm^3 \cdot mm \cdot m^{-2} \cdot d^{-1} \cdot kPa^{-1}$ your **starting** unit or your **target** unit?

thats the starting but i want to go to $cm^3 \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$

26 Feb

Simple one! Only two dimensions change:

Step	Conversion	Factor
1	$cm^3 \rightarrow cm^3$	×1
2	**mm → µm**	×1000 (1 mm = 1000 µm)
3	$m^{-2} \rightarrow m^{-2}$	×1
4	$d^{-1} \rightarrow d^{-1}$	×1
5	**$kPa^{-1} \rightarrow bar^{-1}$**	×100 (1 bar = 100 kPa)

Combined: ×1000 × ×100 = ×100,000 = $\times10^5$

So to convert your values from $cm^3 \cdot mm \cdot m^{-2} \cdot d^{-1} \cdot kPa^{-1}$ to $cm^3 \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$, just **multiply by 10^5**.

correct this please: $m^3 \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$

1. m3 à cm3 (·1)

2. m à µm (·1)
3. m-2 à m-2 (·10-4)
4. s-1 à d-1 (·1/86400)
5. Pa-1à bar-1 (·1/76)

26 Feb

Let's go through each step:

Step 1: m^3 → cm^3 (×1) ✗ 1 m^3 = 10^6 cm^3, not 1.

Correct: m^3 → cm^3, ×10^6

Step 2: m → µm (×1) ✗ Same issue as before — 1 m ≠ 1 µm. 1 m = 10^6 µm.

Correct: m → µm, ×10^6

Step 3: m^{-2} → m^{-2} (×10^{-4}) ✗ You're converting to the same unit but applying a factor — contradicts itself. Also, where are you going? You need to state the **target** unit. What unit do you want for area in the output?

Step 4: s^{-1} → d^{-1} (×1/86400) ✗ The direction is backwards. Going from s^{-1} to d^{-1} means you're converting to a **larger** time unit, so the factor should make the number **smaller** — but 1/86400 would do that. Actually the factor itself is wrong too: 1 d = 86400 s, so **1 s^{-1} = 86400 d^{-1}**, meaning you multiply by **×86400**, not ×1/86400.

Correct: s^{-1} → d^{-1}, ×86400

Step 5: Pa^{-1} → bar^{-1} (×1/76) ✗ 1/76 is wrong here — that's related to the atm/cmHg relationship. The correct factor is: 1 bar = 10^5 Pa, so 1 Pa^{-1} = 10^5 bar^{-1}.

Correct: Pa^{-1} → bar^{-1}, ×10^5

Before I give you the corrected full table —**what is your target unit exactly?** You haven't stated where you want to end up, and step 3 is unclear (m^{-2} staying as m^{-2}?). Once you confirm the full target unit I can give you the clean corrected version.

and this one? $cm^3 \cdot cm \cdot m^{-2} \cdot d^{-1} \cdot atm^{-1}$

1. cm3 à cm3 (·1)
2. cm à µm (·104)
3. m-2 à m-2 (·10-4)
4. d-1 à d-1 (·1)
5. atm-1à bar-1 (·1/76)

26 Feb

What is your target unit? You haven't told me where you're converting **to**, so I can't check if your steps make sense!

the same $cm^3 \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$

26 Feb

Let's go through each step:

Step 1: $cm^3 \rightarrow cm^3$ (×1) ✅

Step 2: cm → μm ($\times 10^4$) ✅ 1 cm = 10^4 μm. Correct.

Step 3: $m^{-2} \rightarrow m^{-2}$ ($\times 10^{-4}$) ❌ You're staying in m^{-2}, so the factor should be ×1. The $\times 10^{-4}$ makes no sense here — nothing is changing.

> **Correct: $m^{-2} \rightarrow m^{-2}$, ×1**

Step 4: $d^{-1} \rightarrow d^{-1}$ (×1) ✅

Step 5: $atm^{-1} \rightarrow bar^{-1}$ (×1/76) ❌ 1/76 is the atm→cmHg relationship, not atm→bar. The correct factor is: 1 atm = 1.01325 bar, so 1 atm^{-1} = (1/1.01325) bar^{-1}.

> **Correct: $atm^{-1} \rightarrow bar^{-1}$, ×1/1.01325 ≈ ×0.9869**

Corrected full calculation:

Step	Conversion	Factor
1	$cm^3 \rightarrow cm^3$	×1
2	cm → μm	$\times 10^4$
3	$m^{-2} \rightarrow m^{-2}$	×1
4	$d^{-1} \rightarrow d^{-1}$	×1
5	$atm^{-1} \rightarrow bar^{-1}$	×1/1.01325 ≈ ×0.9869
Combined		**$\times 10^4$ × 0.9869 ≈ ×9869**

So multiply your values by **~9869** to convert from $cm^3 \cdot cm \cdot m^{-2} \cdot d^{-1} \cdot atm^{-1}$ to $cm^3 \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$.

convert this to g um m-2 d-1 bar -1 $g \cdot mm \cdot m^{-2} \cdot h^{-1} \cdot kPa^{-1}$

1. g à g (·1)
2. mm à µm (·103)
3. m-2 à m-2 (·1)
4. h-1 à d-1 (·1/86400)
5. kPa-1à bar-1 (·1/76)

26 Feb

Let's check each step:

Step 1: g → g (×1) ✅

Step 2: mm → µm ($\times 10^3$) ✅ 1 mm = 1000 µm. Correct.

Step 3: $m^{-2} \rightarrow m^{-2}$ (×1) ✅

Step 4: $h^{-1} \rightarrow d^{-1}$ (×1/86400) ❌ 1/86400 is the conversion from seconds to days, not hours to days. 1 day = 24 hours, so 1 h^{-1} = 24 d^{-1}.

> **Correct: $h^{-1} \rightarrow d^{-1}$, ×24** (because 1 day = 24 hours)

Step 5: $kPa^{-1} \rightarrow bar^{-1}$ (×1/76) ❌ 1/76 is wrong again — that's the atm/cmHg factor. 1 bar = 100 kPa, so 1 kPa^{-1} = (1/100) bar^{-1}.

> **Correct: $kPa^{-1} \rightarrow bar^{-1}$, ×1/100** (because 1 bar = 100 kPa)

Corrected full calculation:

Step	Conversion	Factor
1	g → g	×1
2	mm → µm	$\times 10^3$
3	$m^{-2} \rightarrow m^{-2}$	×1
4	$h^{-1} \rightarrow d^{-1}$	×24
5	$kPa^{-1} \rightarrow bar^{-1}$	×1/100
Combined		**$\times 10^3 \times 24 \times 1/100 = \times 240$**

So multiply your values by **240** to convert from $g \cdot mm \cdot m^{-2} \cdot h^{-1} \cdot kPa^{-1}$ to $g \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$.

can you check this one? $kg \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$

1. kg à g (·10-3)
2. m à µm (·106)
3. m-2 à m-2 (·1)
4. s-1 à d-1 (·86400)
5. Pa-1à bar-1 (·105)

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Let's check each step:

Step 1: kg → g ($\times 10^{-3}$) ✗ You're going from a bigger unit to a smaller one, so the number should get **bigger**, not smaller. 1 kg = 1000 g.

> **Correct: kg → g, ×1000 (= $\times 10^3$)**

Step 2: m → µm ($\times 10^6$) ✓ 1 m = 10^6 µm. Correct.

Step 3: m^{-2} → m^{-2} (×1) ✓

Step 4: s^{-1} → d^{-1} (×86400) ✓ 1 day = 86400 s, so 1 s^{-1} = 86400 d^{-1}. Correct.

Step 5: Pa^{-1} → bar^{-1} ($\times 10^5$) ✓ 1 bar = 10^5 Pa, so 1 Pa^{-1} = 10^5 bar^{-1}. Correct.

Corrected full calculation:

Step	Conversion	Factor
1	kg → g	$\times 10^3$
2	m → µm	$\times 10^6$
3	$m^{-2} \rightarrow m^{-2}$	×1
4	$s^{-1} \rightarrow d^{-1}$	×86400
5	$Pa^{-1} \rightarrow bar^{-1}$	$\times 10^5$
Combined		**$\times 10^3 \times 10^6 \times 86400 \times 10^5 = \times 8.64 \times 10^{16}$**

So multiply your values by **8.64×10^{16}** to convert from $kg \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$ to $g \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$. Only one error — just the direction of the kg→g conversion.

can you calculate the conversion factors? vProperty

Unit Derivation Conversion factor PO2 $cm^3 \cdot mm \cdot m^{-2} \cdot d^{-1} \cdot kPa^{-1}$ 1. cm3 à cm3 (·1) 2. mm à µm (·103) 3. m-2 à m-2 (·1) 4. d-1 à d-1 (·1) 5. kPa-1à bar-1 (·102) $m^3 \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$ 1.

m3 à cm3 (·106) 2. m à µm (·106) 3. m-2 à m-2 (·1) 4. s-1 à d-1 (·8.64·104) 5. Pa-1à bar-1 (·105) OTR $cm^3·m^{-2}·d^{-1}$ 1. cm3 à cm3 (·1) 2. m-2 à m-2 (·10-4) 3. d-1 à d-1 (·86400) 4. ???-1à bar-1 (·1/76) $cm^3·cm·m^{-2}·d^{-1}·atm^{-1}$ 1. cm3 à cm3 (·1) ???? 2. m-2 à m-2 (·10-4) 3. d-1 à d-1 (·1/86400) 4. bar-1à bar-1 (·1/76) PCO2 $cm^3·mm·m^{-2}·d^{-1}·kPa^{-1}$ 1. cm3 à cm3 (·1) 2. mm à µm (·103) 3. m-2 à m-2 (·1) 4. d-1 à d-1 (·1) 5. kPa-1à bar-1 (·102) CO2TR $cm^3·cm·m^{-2}·d^{-1}·atm^{-1}$ 1. cm3 à cm3 (·1) 2. cm à µm (·104) 3. m-2 à m-2 (·1) 4. d-1 à d-1 (·1) 5. atm-1à bar-1 (·9.869·10-1) WVP $g·mm·m^{-2}·h^{-1}·kPa^{-1}$ 1. g à g (·1) 2. mm à µm (·103) 3. m-2 à m-2 (·1) 4. h-1 à d-1 (·24) 5. kPa-1à bar-1 (·102) $g·m^{-1}·s^{-1}·Pa^{-1}$ 1. g à g (·1) 2. ??? à µm (·1) 3. m-2 à m-2 (·10-4) 4. h-1 à d-1 (·1/86400) 5. kPa-1à bar-1 (·1/76) $10^{-8}·g·m^{-1}·s^{-1}·Pa^{-1}$ 1. g
26 Feb

There are several errors and question marks in your derivations. Let me go through everything carefully first, then calculate the correct factors.

Check all derivations and compute correct conversion factors

Here is the full breakdown. I'm assuming your target units are **$cm^3·µm·m^{-2}·d^{-1}·bar^{-1}$** (gas), **$g·µm·m^{-2}·d^{-1}·bar^{-1}$** (WVP), and **$g·m^{-2}·d^{-1}$** (WVTR).

✅ Correct derivations

Property	Unit	CF
PO_2	$cm^3·mm·m^{-2}·d^{-1}·kPa^{-1}$	**$1.00×10^5$**
PO_2	$m^3·m·m^{-2}·s^{-1}·Pa^{-1}$	**$8.64×10^{21}$**
PCO_2	$cm^3·mm·m^{-2}·d^{-1}·kPa^{-1}$	**$1.00×10^5$**
CO_2TR	$cm^3·cm·m^{-2}·d^{-1}·atm^{-1}$	**$9.869×10^3$**
WVP	$g·mm·m^{-2}·h^{-1}·kPa^{-1}$	**$2.40×10^6$**
WVP	$cm^3·mm·m^{-2}·d^{-1}·kPa^{-1}$	**$1.00×10^5$**
WVP	$kg·m·m^{-2}·s^{-1}·Pa^{-1}$	**$8.64×10^{18}$**
WVTR	$g·m^{-2}·d^{-1}$	**1**
WVTR	$g·m^{-2}·s^{-1}$	**$8.64×10^4$**

❌ Errors found

OTR: $cm^3·m^{-2}·d^{-1}$ — Steps 2, 3, 4 are all wrong. This unit has no thickness or pressure dimension, so steps 2–4 don't apply at all. If the target is also $cm^3·m^{-2}·d^{-1}$ then **CF = 1**. You need to clarify what target unit you want for OTR.

CO_2TR (first entry): $cm^3·cm·m^{-2}·d^{-1}·atm^{-1}$ — Step 2 (cm→µm) is completely missing. Steps 3, 4, 5 are all wrong ($×10^{-4}$, ×1/86400, ×1/76 should all be ×1, ×1, ×0.9869). Correct CF = **$9.869×10^3$**.

WVP: $g·m^{-1}·s^{-1}·Pa^{-1}$ — The key insight you're missing is that $g·m^1 = g·m·m^{-2}$, so **m is the thickness dimension**, not a separate step. Correct steps: m→µm $×10^6$, m^{-2}→m^{-2} ×1, s^{-1}→d^{-1} ×86400 (not ÷86400!), Pa^{-1}→bar^{-1} $×10^5$ (not ×1/76). Correct CF = **$8.64×10^{15}$**.

WVP: $10^{-8}·g·m^{-1}·s^{-1}·Pa^{-1}$ — Same errors as above, plus step 6 should be $×10^8$ (to cancel the 10^{-8} prefix), not $×10^{10}$. Correct CF = **$8.64×10^{23}$**.

WVP: 10^{-7}·g·m^{-1}·d^{-1}·Pa^{-1} — Same pattern: m→μm ×10^6, m^{-2}→m^{-2} ×1, d^{-1}→d^{-1} ×1, Pa^{-1}→bar^{-1} ×10^5, and step 6 should be ×10^7 not ×10^{10}. Correct CF = **1.00×10^{18}**.

The two most common mistakes throughout: using ×1/76 for pressure (that's the atm→cmHg factor, not relevant here), and getting the direction of the time conversion backwards (dividing when you should multiply).

can you give me a table with the OTR values of this table in cm^3·m^{-2}·d^{-1}·bar^{-1} . SamplesAverage thicknessTensile strengthElongation at breakElastic modulusOTR @ 20 % RHOTR @ 40 % RHWVTR @ 40 % RHEmpty CellμmMPa%MPacm3/(m2.24h)g/(m2.24h.mmHg)CP film114.20 ± 5.018.07 ± 2.4516.74 ± 3.46159.84 ± 36.240.23 ± 0.07 @ 21 % Oxygen 1.15 ± 0.04 @ 100 % Oxygen1.34 ± 0.16 @ 21 % Oxygen 5.77 ± 1.14 @ 100 % Oxygen26.34 ± 2.16MCP film119.83 ± 14.133.57 ± 0.565.66 ± 1.69122.61 ± 18.610.19 ± 0.05 @ 21 % Oxygen 0.94 ± 0.04 @ 100 % Oxygen0.96 ± 0.11 @ 21 % Oxygen 4.58 ± 1.14 @ 100 % Oxygen17.80 ± 1.24. these are the methods: 2.5.3. Oxygen and water vapour barrier properties All polymeric films' oxygen and water vapour barrier properties were investigated on a 50 cm2 surface sample using a TotalPerm permeability analyser (ExtraSolution®Srl, Capannori, Italy) based on the isostatic method and equipped with a double sensor system, that is, an electrochemical sensor for oxygen and an infrared sensor for water vapour. The oxygen transmission rate (OTR, in cm3 m−2 day−1) was determined at 23 °C and 50 % relative humidity (RH), according to the standard methods ASTM F2476 and ASTM F2622, respectively, with a carrier flow (N2) of 10 mL min−1 and a one-atmosphere analyte partial pressure difference between the specimen's two sides. For OTR, values at a 0.21 partial pressure difference were also calculated to simulate the oxygen barrier performance under natural (atmospheric) conditions. The water vapour transmission rate (WVTR, in g m−2 day−1) was measured according to the ASTM F1249 standard method, with a carrier flow (N2) of 10 mL min−1 at 38 °C and 90 % RH (tropical conditions). All the analyses were conducted with each sample's external side (i.e., the external environment) facing the upper semi-chamber, where the humid test gas (i.e., oxygen) was flushed [19].

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Looking at this carefully — the OTR values in the table are in**cm^3/(m^2·24h)**, and the test conditions tell us the pressure difference is **1 atm** (one-atmosphere analyte partial pressure difference).

So the starting unit is effectively **cm^3·m^{-2}·d^{-1}·atm^{-1}** (since 24h = 1 day), and we need to convert to **cm^3·m^{-2}·d^{-1}·bar^{-1}**.

Only **one dimension changes**: atm → bar.

1 atm = 1.01325 bar → atm^{-1} → bar^{-1}: ×(1/1.01325) = ×0.9869

Convert OTR values from atm to bar

Here are the converted values:

Sample	Condition	OTR (cm^3·m^{-2}·d^{-1}·atm^{-1})	OTR (cm^3·m^{-2}·d^{-1}·bar^{-1})
CP film	21% O_2 @ 20% RH	0.23 ± 0.07	0.2270 ± 0.0691
CP film	100% O_2 @ 20% RH	1.15 ± 0.04	1.1350 ± 0.0395
CP film	21% O_2 @ 40% RH	1.34 ± 0.16	1.3225 ± 0.1579

Sample	Condition	OTR ($cm^3 \cdot m^{-2} \cdot d^{-1} \cdot atm^{-1}$)	OTR ($cm^3 \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$)
CP film	100% O_2 @ 40% RH	5.77 ± 1.14	5.6945 ± 1.1251
MCP film	21% O_2 @ 20% RH	0.19 ± 0.05	0.1875 ± 0.0493
MCP film	100% O_2 @ 20% RH	0.94 ± 0.04	0.9277 ± 0.0395
MCP film	21% O_2 @ 40% RH	0.96 ± 0.11	0.9474 ± 0.1086
MCP film	100% O_2 @ 40% RH	4.58 ± 1.14	4.5201 ± 1.1251

CF = ×0.9869 (only atm→bar changes, everything else stays the same)

Since 1 atm ≈ 1.013 bar, the values barely change — less than 1.3% difference. Worth noting that the table header says **$cm^3/(m^2 \cdot 24h)$** which confirms 24h = 1 day, so no time conversion is needed either.

can you fix this conversion factor? $cm^3 \cdot m^{-2} \cdot d^{-1} \cdot 0.1mPa^{-1}$

1. cm3 à cm3 (·1)
2. m-2 à m-2 (·10-4)
3. d-1 à d-1 (·86400)
4. 0.1mPa-1à bar-1 (·1/76)

to cm 3 m-2 d-1 bar-1 please

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Let's go through each step:

Step 1: cm^3 → cm^3 (×1) ✅

Step 2: m^{-2} → m^{-2} ($\times 10^{-4}$) ❌ Target is also m^{-2}, nothing changes. ×1.

Step 3: d^{-1} → d^{-1} (×86400) ❌ Target is also d^{-1}, nothing changes. ×1.

Step 4: 0.1 MPa^{-1} → bar^{-1} (×1/76) ❌ 1/76 is wrong (that's atm→cmHg). The correct conversion: 1 bar = 0.1 MPa, so 1 $(0.1\ MPa)^{-1}$ = 1 bar^{-1} → **×1**.

Corrected steps:

Step	Conversion	Factor
1	cm^3 → cm^3	×1
2	m^{-2} → m^{-2}	×1
3	d^{-1} → d^{-1}	×1
4	0.1 MPa^{-1} → bar^{-1}	×1
Combined		**×1**

CF = 1 — because 0.1 MPa and bar are actually the same pressure (1 bar = 0.1 MPa exactly), so no conversion is needed at all. The units are identical in disguise!

fix these and calculate thje conversion factor please: $m^3 \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$

1. m^3 à g (·1)
2. m à µm (·1)
3. m-2 à m-2 (·10-4)
4. s-1 à d-1 (·1/86400)
5. Pa-1à bar-1 (·1/76) $10^{-8} \cdot g \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$
6. g à g (·1)
7. m à µm (·1)
8. m-2 à m-2 (·10-4)
9. s-1 à d-1 (·1/86400)
10. Pa-1 à bar-1 (·1/76) 6.·10-8 à ·1 (·1010)

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Let me go through both carefully. I need to know your target unit first — based on the context I'll assume **$g \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$**.

Unit 1: $m^3 \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1}$

Step 1: m^3 → g (×1) ✗ Two problems. First, m^3 → g is a mass/volume conversion, which only makes sense for liquid water (density = 1 g/cm^3 = 10^6 g/m^3). But even then ×1 is wrong — 1 m^3 of water = 10^6 g. Second, the target unit is **g**, but actually m^3 here is a **gas volume**, not liquid water, so this conversion doesn't make sense at all. You should convert m^3 → cm^3 first if going to a volumetric unit, or the unit itself may be wrong.

Assuming target is $g \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$ and m^3 means m^3 of gas (not liquid):

> **m^3 → cm^3: ×10^6** (1 m^3 = 10^6 cm^3) — but then unit is cm^3 not g

If you truly want grams, you'd need a molar mass conversion which depends on the gas. **Please clarify what gas this is.**

Step 2: m → µm (×1) ✗ 1 m = 10^6 µm, so the factor should be ×10^6.

> **Correct: ×10^6**

Step 3: m^{-2} → m^{-2} (×10^{-4}) ✗ Target is also m^{-2}, nothing changes. ×1.

> **Correct: ×1**

Step 4: s^{-1} → d^{-1} (×1/86400) ✗ Going from per second to per day — since a day is longer, the number gets **bigger**. 1 s^{-1} = 86400 d^{-1}.

> **Correct: ×86400**

Step 5: Pa^{-1} → bar^{-1} (×1/76) ✗ 1/76 is atm→cmHg, completely wrong here. 1 bar = 10^5 Pa → Pa^{-1} → bar^{-1}:

$\times 10^5$.

Correct: $\times 10^5$

Unit 2: $10^{-8}\cdot g\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$

Step 1: g → g (×1) ✅

Step 2: m → μm (×1) ❌ Same as above. 1 m = 10^6 μm.

Correct: $\times 10^6$

Step 3: m^{-2} → m^{-2} ($\times 10^{-4}$) ❌ $m\cdot m^{-2} = m^{-1}$, so the m (thickness) and m^{-2} (area) are **already combined** into m^{-1}. You should not apply a separate area correction — step 2 converting m→μm already handles everything.

Correct: ×1

Step 4: s^{-1} → d^{-1} (×1/86400) ❌ Wrong direction again. ×86400.

Correct: ×86400

Step 5: Pa^{-1} → bar^{-1} (×1/76) ❌ Same error. 1 bar = 10^5 Pa → $\times 10^5$.

Correct: $\times 10^5$

Step 6: 10^{-8} → 1 ($\times 10^{10}$) ❌ To cancel a 10^{-8} prefix you multiply by $\times 10^8$, not $\times 10^{10}$.

Correct: $\times 10^8$

Corrected steps and CFs:

Calculate corrected conversion factors for both units

Unit 1: $m^3\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$ → $cm^3\cdot \mu m\cdot m^{-2}\cdot d^{-1}\cdot bar^{-1}$

Step	Conversion	Your factor	Correct factor
1	$m^3 \to cm^3$	×1 ❌	$\times 10^6$
2	m → μm	×1 ❌	$\times 10^6$
3	$m^{-2} \to m^{-2}$	$\times 10^{-4}$ ❌	×1
4	$s^{-1} \to d^{-1}$	×1/86400 ❌	×86400
5	$Pa^{-1} \to bar^{-1}$	×1/76 ❌	$\times 10^5$
CF			**8.64×10^{21}**

⚠ Also: m^3 → g makes no sense for a gas — you can only convert volume to mass if you know the molar mass. Since this is likely a gas permeability unit, $m^3 \to cm^3$ ($\times 10^6$) is the right step.

Unit 2: $10^{-8}\cdot g\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$ → $g\cdot \mu m\cdot m^{-2}\cdot d^{-1}\cdot bar^{-1}$

Step	Conversion	Your factor	Correct factor

Step	Conversion	Your factor	Correct factor
1	g → g	×1 ✓	×1
2	m → μm	×1 ✗	$\times 10^{6}$
3	$m^{-2} \rightarrow m^{-2}$	$\times 10^{-4}$ ✗	×1
4	$s^{-1} \rightarrow d^{-1}$	×1/86400 ✗	×86400
5	$Pa^{-1} \rightarrow bar^{-1}$	×1/76 ✗	$\times 10^{5}$
6	cancel 10^{-8}	$\times 10^{10}$ ✗	$\times 10^{8}$
CF			**8.64×10^{23}**

The **same four mistakes** appear in both units: ×1 instead of $\times 10^{6}$ for the length, $\times 10^{-4}$ instead of ×1 for area, wrong direction on time, and ×1/76 instead of $\times 10^{5}$ for pressure. The ×1/76 error keeps recurring — please remember that **1/76 is atm→cmHg only**, it has nothing to do with Pa→bar.

is this correct? $m^{3}\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$

1. m^{3} à g (·106)
2. m à μm (·106)
3. m-2 à m-2 (·1)
4. s-1 à d-1 (·8.64·104)
5. Pa-1à bar-1 (·105) $10^{-8}\cdot g\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$
6. g à g (·1)
7. m à μm (·106)
8. m-2 à m-2 (·1)

26 Feb

Almost perfect! Just one issue:

Unit 1: $m^{3}\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$

Step 1: m^{3} → g ($\times 10^{6}$) ✗

The factor $\times 10^{6}$ is correct **only if** m^{3} means liquid water (1 m^{3} water = 10^{6} g). But m^{3} in a permeability unit is a **gas volume**, not liquid water, so you cannot convert it to grams without knowing the molar mass of the gas. The correct conversion is:

> **m^{3} → cm^{3}: $\times 10^{6}$** → giving output unit **cm^{3}**$\cdot\mu m\cdot m^{-2}\cdot d^{-1}\cdot bar^{-1}$

Steps 2–5 are all ✓ correct.

CF = $10^{6} \times 10^{6} \times 1 \times 8.64\times 10^{4} \times 10^{5} = 8.64\times 10^{21}$

Unit 2: $10^{-8}\cdot g\cdot m\cdot m^{-2}\cdot s^{-1}\cdot Pa^{-1}$

All 6 steps ✓ **completely correct!**

CF = $1 \times 10^6 \times 1 \times 8.64\times10^4 \times 10^5 \times 10^8 = 8.64\times10^{23}$

So unit 2 is perfect. Unit 1 just has a labelling issue in step 1 — the factor $\times10^6$ is right but write it as **m^3 → cm^3** not m^3 → g.

i am talking about wvp

26 Feb

In that case for WVP, m^3 of **liquid water** does make sense! And 1 m^3 water = 10^6 g (since density of water = 1 g/cm^3 = 10^6 g/m^3), so:

Unit 1: $m^3 \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1} \rightarrow g \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$

All steps ✅ **completely correct!**

CF = $10^6 \times 10^6 \times 1 \times 8.64\times10^4 \times 10^5 = 8.64\times10^{21}$

Unit 2: $10^{-8} \cdot g \cdot m \cdot m^{-2} \cdot s^{-1} \cdot Pa^{-1} \rightarrow g \cdot \mu m \cdot m^{-2} \cdot d^{-1} \cdot bar^{-1}$

All steps ✅ **completely correct!**

CF = $1 \times 10^6 \times 1 \times 8.64\times10^4 \times 10^5 \times 10^8 = 8.64\times10^{23}$

Both derivations are fully correct. My apologies for the confusion in the previous message — I should have asked about the context first. For WVP, converting m^3 liquid water to grams with $\times10^6$ is perfectly valid.